ESE 3123 One-Shot Notes

Thermo-Fluid Devices · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the teachers' material (Fahim sir, Shameem sir) and the textbooks they follow (Cengel & Cimbala, Bansal, Incropera, Holman)

One-shot notes: ESE 3123 Thermo-Fluid Devices

Pipe flow (A1)

Re = ρVD/μ = VD/ν: laminar < 2300, transition 2300-4000, turbulent > 4000. Entrance length $L_h \approx$ 0.05ReD (laminar), 10D (turbulent); beyond it the flow is fully developed.

Darcy-Weisbach $h_L = f(L/D)(V^2/2g)$; laminar f = 64/Re; turbulent from Moody/Colebrook (f depends on Re and ε/D). Minor loss $h = K\,V^2/2g$.

Laminar profile: $u = u_{max}(1 - r^2/R^2)$, $V_{avg} = u_{max}/2$, $\Delta P = 32\mu LV/D^2$ (Hagen-Poiseuille). Turbulent profile is flatter.

Pipe numericals (A2)

Steps: Re → f → h_L → $\Delta P = \rho g h_L$ → $W = Q\Delta P$.

2019 2b: Re 1777, f 0.036, h_L 4.46 m, ΔP 43.7 kPa, W 0.28 W. 2023 6d: Re 1836, ΔP 188.2 kPa, h_L 19.2 m, W 0.71 W.

Series: same Q, h_L add. Parallel: same h_L, Q add.

2021 1c ratio 45.7. 3-inch pipe: 10S ID 3.260 in; sch 40 ID 3.068 in; XXS ID 2.300 in.

Dimensional analysis (A3)

Fundamental dimensions M, L, T (θ). Homogeneity: every term has the same dimensions. Rayleigh: write $Q = kA^aB^b\ldots$, equate powers of M, L, T.

Buckingham π: n variables, m dimensions ⇒ (n − m) π groups; choose m repeating variables (geometric, kinematic, dynamic: usually D, V, ρ), each $\pi = D^aV^b\rho^cX$. Drag gives $F/(\rho V^2D^2) = f(Re)$.

Flow meters (A4)

Venturi: $Q = C_d a_1 a_2 \sqrt{(2gh)}/\sqrt{(a_1^2 - a_2^2)}$, $h = x(S_m/S - 1)$; C_d 0.95-0.99. Orifice same with C_d 0.6-0.65 (vena contracta). Pitot $V = C_v\sqrt{(2gh)}$.

Rotameter: float rises in tapered tube until weight = drag + buoyancy (variable area). Turbine meter: rotor speed ∝ Q.

Venturi Q 125.8 L/s. Pitot 2020: h 2.451 m, V 6.93 m/s; 2022: 2.328 m, 6.76 m/s. Air venturi 2019 6d: 3.12 kg/s (3.25 if incompressible).

Centrifugal pump (A5)

Parts: impeller, casing (volute, vortex, diffuser), suction pipe with foot valve and strainer, delivery pipe with valve. Converts mechanical energy into pressure energy by centrifugal action.

Work per unit weight = $V_{w2}u_2/g$ (radial entry, V_{w1} = 0). Manometric head $H_m = h_s + h_d + h_{fs} + h_{fd} + V_d^2/2g$.

$\eta_{mano} = gH_m/(V_{w2}u_2)$; η_{mech} = impeller power/shaft power; $\eta_o = \rho gQH_m/P$.

Pump performance (A6)

Curves: H, P, η vs Q at constant N; BEP at peak η. Affinity: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$.

$N_s = N\sqrt{Q}/H^{3/4}$ (per stage, per eye). Cavitation: local $P < P_{vap}$ ⇒ bubbles collapse, pitting, noise, drop of H and η. $NPSH_A = (P_{atm} - P_v)/\rho g - h_s - h_{fs} \geq NPSH_R$.

Priming: fill suction pipe and casing with liquid before starting (air gives too little head).

Pump numericals (A7)

Series: H adds. 2020 3d: 31.9 m per pump ⇒ 3.14 ⇒ 4 pumps. Multistage: $n = H/H_{stage}$; 2021 3d: N_s 322, 18.8 m/stage, 10 stages, D 58.7 cm. 2022 4d: N_s 672.7, 19 stages, D 36.7 cm (alt. 10 stages, 35.8 cm). Affinity: 2023 3d: 866 rpm, 92.4 gpm, 9.24 hp. 2019 7d: 1072 rpm, 122.5 gpm, 14.7 hp.

Reciprocating pump (A8)

Q_{th} = ALN/60 (single acting), 2ALN/60 (double). Slip = $Q_{th} - Q_{act}$; % slip = $(1 - C_d)100$; negative slip when long suction, high speed. $P = \rho gQ_{th}(h_s + h_d)$.

2020 2b: C_d 0.960, slip 3.96%, P 4.21 kW. 2021 4d: slip −1.86%, C_d 1.019, P 7.38 kW. 2023 4d: slip 8.33%, C_d 0.917. 2019 8c: N 83.2 rpm, H 117.1 m, P 130.2 kW.

Plunger: piston-type, high pressure. Diaphragm: flexible membrane, leak-free, slurries/chemicals.

Compressors and fans (A9)

Fan (pressure ratio < 1.1), blower (1.1-1.2), compressor (> 1.2). Reciprocating (high P, low Q), rotary screw/vane, centrifugal and axial (large Q).

Isothermal efficiency = isothermal work/actual work; FD fan pushes air into the furnace, ID fan pulls gas out through the stack.

Compressible basics (B1)

Ma = V/c; incompressible < 0.3, subsonic < 1, sonic = 1, transonic 0.8-1.2, supersonic 1-5, hypersonic > 5.

$c = \sqrt{(K/\rho)} = \sqrt{(kRT)}$; air 300 K: 347 m/s. $T_0/T = 1 + (k - 1)Ma^2/2$; $P_0/P = (T_0/T)^{k/(k-1)}$. Adiabatic Bernoulli: $[k/(k-1)]p/\rho g + V^2/2g + z = C$.

Air critical: T^*/T_0 0.8333, P^*/P_0 0.5283, ρ^*/ρ_0 0.6339. 2021 5d: 324.8 m/s vs 291.4 m/s incompressible.

Nozzles (B2)

$dA/A = (Ma^2 - 1)dV/V$: subsonic nozzle converges, supersonic nozzle diverges, Ma = 1 only at the throat.

$\dot{m}_{max} = A^*P_0\sqrt{(k/RT_0)}[2/(k + 1)]^{(k+1)/2(k-1)} = 0.6847A^*P_0/\sqrt{(RT_0)}$ air. Back pressure below P* ⇒ choked, no further rise in $\dot{m}$.

2022 5c: T* 301 K, P* 118 kPa, A^*/A_1 0.629. 2020 4c CO_2: V 164.9, ρ 13.9, A 13.1 cm^2, Ma 0.494. 2023 1d: $\dot{m}$ 2.86 kg/s, V_e 845 m/s, A_e 33.75 cm^2. 2019 5d: 6.78 and 7.11 kg/s.

Shocks (B3)

Normal shock: supersonic → subsonic, P, T, ρ, s up, P_0 down, T_0 same. Oblique: inclined, flow turned, weaker.

$Ma_2^2 = (Ma_1^2 + 5)/(7Ma_1^2 - 1)$ (air); $P_2/P_1 = (1 + kMa_1^2)/(1 + kMa_2^2)$ (Rayleigh line). Shock states = Rayleigh ∩ Fanno.

Ma 2: Ma_2 0.577, P ratio 4.5, T ratio 1.6875. Ma 2.5: 0.513, 7.125, 2.1375. Ma 2.2: 0.547, 5.48, 1.857.

HX fundamentals (B4)

Recuperator, regenerator, direct contact; parallel, counter, cross flow; double pipe, shell-and-tube, plate, compact.

$1/UA = 1/h_iA_i + R_{f,i}/A_i + \ln(D_o/D_i)/2\pi kL + R_{f,o}/A_o + 1/h_oA_o$; thin clean: $U = 1/(1/h_i + 1/h_o)$. $R_f = 1/U_{dirty} - 1/U_{clean}$.

$Q = UA\Delta T_{lm}$, $\Delta T_{lm} = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$.

Counter flow: bigger LMTD, $T_{c,o}$ can exceed $T_{h,o}$.

LMTD numericals (B5)

Energy balance first, then ΔT_{lm}, $A = Q/U\Delta T_{lm}$, $L = A/\pi D$.

2020 5c: 301 kW, LMTD 92.0, L 108.5 m. 2020 6b: 28.4 m^2. 2022 7c: 15.8 m^2. 2021 7c: U 43.7, L 92.5 m. 2023 7c: L 0.968 m; R_f 9.5×10^{-4}. 2019 4c: U 37.7, L 66.6 m. 2023 8c: $T_{w,o}$ 18.5 °C, UA 785 W/K, L 5.0 m.

Effectiveness-NTU (B6)

$\varepsilon = Q/[C_{min}(T_{h,i} - T_{c,i})]$; $NTU = UA/C_{min}$; $C_r = C_{min}/C_{max}$.

Parallel: $\varepsilon = [1 - e^{-NTU(1+C_r)}]/(1 + C_r)$. Counter: $\varepsilon = [1 - e^{-NTU(1-C_r)}]/[1 - C_re^{-NTU(1-C_r)}]$; $NTU = [1/(C_r - 1)]\ln[(\varepsilon - 1)/(\varepsilon C_r - 1)]$.

2021 8c: $\dot{m}_w$ 0.403 kg/s, ε 0.75, NTU 2.35, A 2.18 m^2. 2022 8d: NTU 2.50, ε 0.8325, Q 133.3 kW, $T_{c,o}$ 86.6 °C.

Definitions to know by heart

Cavitation, **priming**, **NPSH**, **specific speed** (speed of a geometrically similar pump giving 1 m^3/s against 1 m), **slip**, **manometric efficiency**, **vena contracta**, **C_d**, **Mach number**, **stagnation property**, **choking**, **shock wave**, **fouling factor**, **effectiveness**, **NTU**, **heat capacity rate**.

Sketches to practise

1 Reynolds regimes and Moody chart; 2 entrance region; 3 laminar vs turbulent profile; 4 venturimeter; 5 orifice with vena contracta; 6 Pitot tube; 7 rotameter; 8 centrifugal pump and casings; 9 velocity triangles; 10 pump curves; 11 NPSH/suction; 12 reciprocating pump and indicator diagram; 13 C-D nozzle pressure plot; 14 back-pressure effect; 15 normal shock and Rayleigh-Fanno; 16 parallel vs counter temperature profiles; 17 ε-NTU chart.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
AODD	air-operated double diaphragm (pump)
ASME	American Society of Mechanical Engineers
BEP	best efficiency point
BHP	brake horsepower
C-D	converging-diverging
FD	forced draught
gpm	gallons per minute
hp	horsepower
HX	heat exchanger
ID	induced draught (or inside diameter)
kPa	kilopascal
LMTD	log mean temperature difference
MPa	megapascal
NPSH	net positive suction head
NTU	number of transfer units
rpm	revolutions per minute
TEMA	Tubular Exchanger Manufacturers Association
WHP	water horsepower
XXS	double extra strong (pipe schedule)